Math Workout

Variables and calc

A rectangle is 12 cm wide and 5 cm tall. Area: 12 . 5 = 60 cm².

Double of 3.5 is 7.

Thousands still work: 100001.

Fractions, roots, powers

$\frac{2}{3}$

$\frac{x+1}{x-1}$

x^2

$(x+1)^n$

$\sqrt[2]{\{x\}}$

$\sqrt[3]{\{x+1\}}$

$\sqrt[3]{\{27\}}$

$\sqrt{16}$

$|x-5|$

Nested: $\frac{\sqrt{4}}{2^3}$

Trigonometry and logarithms

$\sin(x)$

$\cos(x)$

$\tan(x)$

$\cot(x)$

$\sec(x)$

$\csc(x)$

$\arcsin(x)$

$\arccos(x)$

$\arctan(x)$

$\sinh(x)$

$\cosh(x)$

$\tanh(x)$

$\coth(x)$

$\log(10)$

$\ln(2.718)$

Sums, products, limits

$$\sum_{i=1}^{n}(i)$$

$$\prod_{i=1}^{n}(i)$$

$\lim_{x\to 0}\left(\frac{\sin(x)}{x}\right)$

Integrals

$\int x^2$

$\int x^2 \,\mathrm{d}x$

$\int_0^1 x^2$

$\int_0^1 x^2 \,\mathrm{d}x$

$\oint_C^F l$

$\int_0^1 x^2$

Piecewise functions and equation systems

$$\begin{cases} x+y=3 \\ x-y=1 \end{cases}$$

$$\begin{cases} xx>0 \\ 0x\leq 0 \end{cases}$$

$$\begin{cases} xx>0 \\ 0x\leq 0 \end{cases}$$

Vectors and binomials

$$\binom{3}{4}$$

$$\binom{-2}{5}$$

$$\binom{12}{6}$$

$$\binom{n}{k}$$

$$\binom{5}{2}$$

Matrices and tables

$$\begin{pmatrix} 1 & 2 \\ 3 & 4 \end{pmatrix}$$

$$\begin{pmatrix} a & b \\ c & d \end{pmatrix}$$

Name	Age
Alice	20
Bob	22

Inline math and subscripts

The solution is $x = 2$.

Discriminant: $b^2 - 4ac$

Corners A_1 and x_n, vector $\binom{3}{4}$.

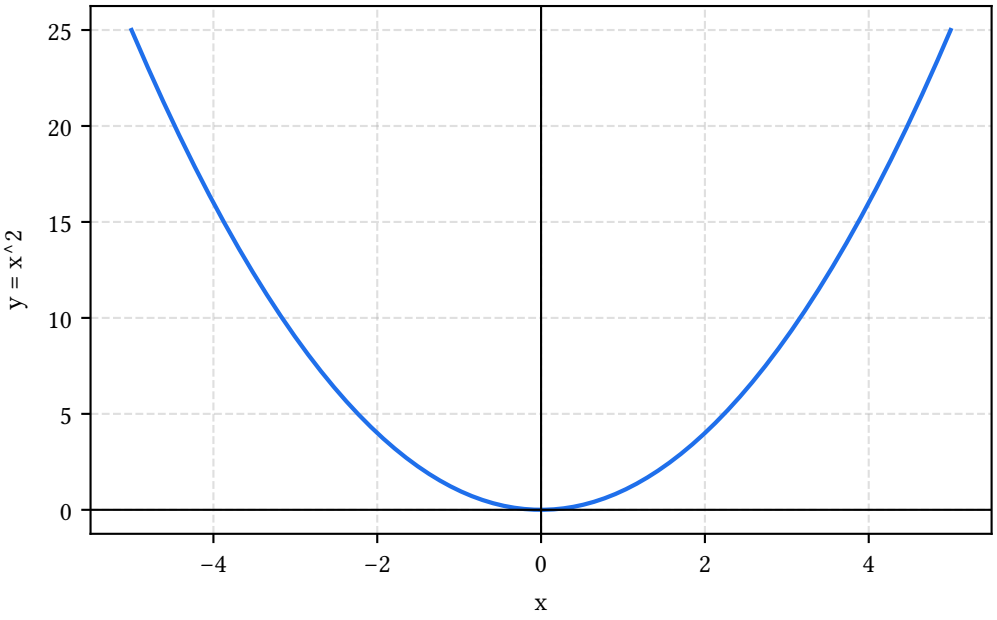
Spacing and symbols

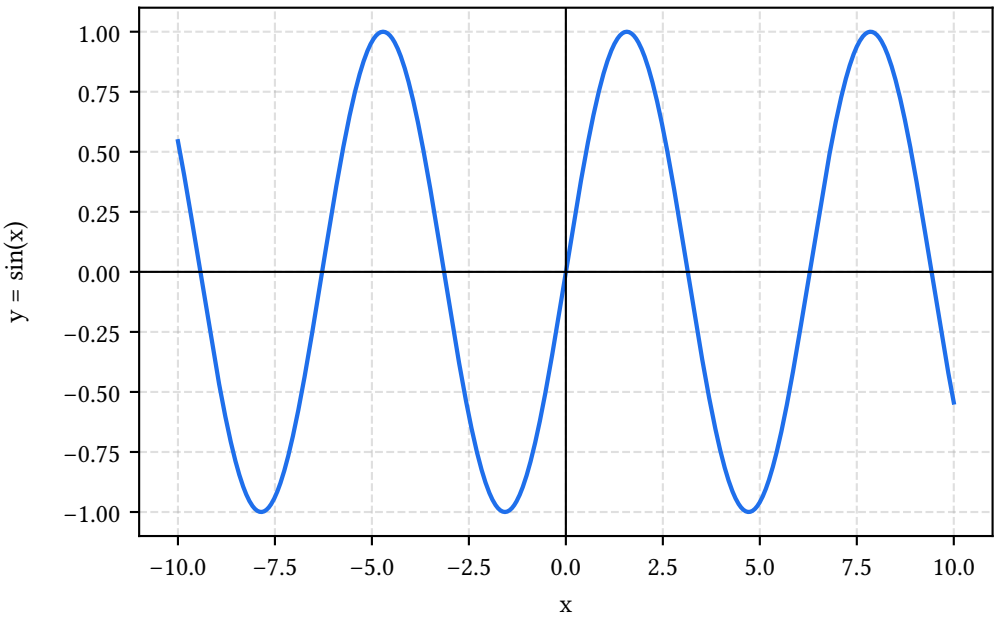
a quad b quad quad c thin d med e

$\alpha \beta \gamma \pi \infty$

If $x \geq 0$ and $x \leq 10$ then $x \neq 5 \implies$ done.

Plots





Everything combined

$$\frac{\int_0^3 x^2}{\binom{4}{2}}$$